

Faculty Teaching Handbook

M.Sc. Mathematics with Artificial Intelligence & Data Science

Lecture 1 (Part 1)

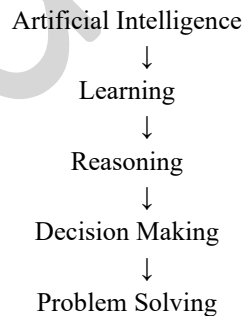
Introduction to Artificial Intelligence – "AI is Already Around Us"

Learning Outcomes

- Define Intelligence and Artificial Intelligence.
- Explain why AI is one of the most influential technologies of the 21st century.
- Identify AI applications in daily life.
- Differentiate conventional programming from AI-enabled systems.
- Develop curiosity about Mathematics behind AI.

- *How many of you have used Google Maps today?*
- *How many watched YouTube today?*
- *How many used ChatGPT, Instagram or WhatsApp?*
- *How many unlocked your phone using Face Unlock or Fingerprint?*

Most people think AI means robots. That is a myth. AI is already inside your phone, banking app, shopping app, social media, navigation system and healthcare applications. During this semester, we will not simply study AI—we will understand how intelligent systems think, learn and make decisions. We will also discover how Mathematics is the foundation of Artificial Intelligence. By the end of this course, whenever you use ChatGPT, Google Maps or Netflix, you will understand the mathematical ideas and algorithms working behind them.

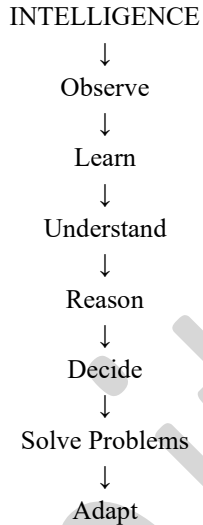


Lecture 1 – Part 2

Understanding Intelligence and Artificial Intelligence

Students, in the previous discussion we saw that AI is already around us. Before defining Artificial Intelligence, let us answer one important question: "What is Intelligence?" Without understanding intelligence, we cannot understand AI.

"Who is an intelligent person?"



What is Intelligence?

Intelligence is the ability to acquire knowledge, understand situations, learn from experience, reason logically, solve problems, adapt to new situations and make appropriate decisions.

Characteristics of Intelligence

- Learning
- Reasoning
- Problem Solving
- Decision Making
- Adaptation
- Creativity

What is Artificial Intelligence?

Artificial = Man-made

Intelligence = Ability to Think, Learn and Decide

Artificial Intelligence is the branch of Computer Science that develops intelligent systems capable of learning from data, reasoning, solving problems and making decisions similar to humans.

Simple Definition

Artificial Intelligence is the science of making machines intelligent enough to perform tasks that normally require human intelligence.

Traditional Programming vs AI

Traditional Programming	Artificial Intelligence
Fixed instructions	Learns from data
Same output	Improves with experience
Rule-based	Data-driven
Cannot learn	Learns continuously

Traditional Programming *Input → Program → Output*

Artificial Intelligence *Data → Learning → Model → Prediction → Decision*

Mathematics Connection

Artificial Intelligence = Mathematics + Algorithms + Data

Core mathematical foundations include Logic, Linear Algebra, Probability, Statistics and Optimization.

Lecture 1 – Part 3

AI Around Us: Real-Life Applications, Stories and Classroom Activities

"Imagine you wake up at 6:00 AM. Your phone unlocks using Face Unlock. Google Maps suggests the fastest route. You pay using UPI, watch YouTube, receive shopping recommendations from Amazon, ask ChatGPT a question and end the day watching Netflix recommendations. Without realizing it, you have used Artificial Intelligence dozens of times in a single day."

'Was AI only inside robots? Or has it already become a part of your life?'

"Where did you use AI today?" Ask every student to mention one example. Do not evaluate immediately; encourage participation.

Morning → Mobile → AI
Travel → Google Maps → AI
Shopping → Amazon → AI
Learning → ChatGPT → AI
Entertainment → Netflix/YouTube → AI

25 Real-Life Examples of AI

Application	AI Role
Google Maps	Shortest route prediction
ChatGPT	Question answering and text generation
YouTube	Video recommendation
Netflix	Movie recommendation
Amazon	Personalized shopping
Flipkart	Product recommendation
Instagram	Feed ranking
Facebook	Friend suggestions
LinkedIn	Job recommendation
Spotify	Music recommendation
Google Translate	Language translation

Gmail	Spam filtering
Face Unlock	Face recognition
Phone Camera	Scene detection
Alexa	Voice assistant
Siri	Speech recognition
Swiggy	Delivery optimization
Uber/Ola	Demand prediction
UPI Fraud Detection	Fraud detection
Banking	Credit scoring
Healthcare	Disease prediction
Agriculture	Crop monitoring
Smart Watch	Health monitoring
CCTV	Facial recognition
ISRO Satellite Images	Image analysis

Connect AI with Mathematics:

- Google Maps → Graph Theory
- Recommendation Systems → Probability & Statistics
- ChatGPT → Linear Algebra
- Fraud Detection → Statistics
- Image Recognition → Matrix Operations
- Route Optimization → Optimization Techniques

Activity: Divide the class into five groups. Give each group three minutes to list as many AI applications as they can find on their smartphones. Ask each group to explain one application and identify what makes it intelligent.

Mini Case Study

Case: One day all AI systems stop working. Discuss what will happen to banking, transport, healthcare, education, communication and social media. This activity helps students appreciate the importance of AI.

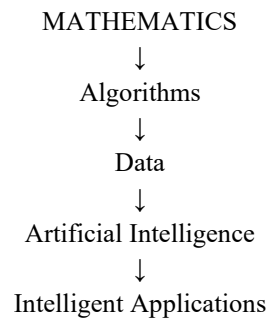
Quick Quiz

1. What is the difference between automation and AI?
2. Is Google Maps an AI application? Why?
3. Give five AI applications used every day.
4. Which AI application do you use most frequently?
5. Why is ChatGpt considered an AI system?

Homework

Prepare a one-page report titled 'Artificial Intelligence in My Daily Life'. Identify at least 20 AI applications you use in one day and explain the purpose of each in one sentence.

'Many students believe AI is only programming. That is a myth. Programming is the tool, but Mathematics is the language of Artificial Intelligence. Without Mathematics, modern AI cannot exist.'



Why Mathematics Matters in AI

Artificial Intelligence learns patterns from data. Learning requires mathematical models, calculations and optimization. Every modern AI system—from Google Maps to ChatGPT—depends on mathematical concepts.

Core Mathematical Foundations

Mathematical Area	Role in AI	Example
Logic	Used for reasoning, inference, expert systems and decision making.	Medical expert system
Probability	Handles uncertainty, prediction, spam filtering and risk estimation.	Weather prediction
Linear Algebra	Represents data as vectors and matrices; fundamental for machine learning and deep learning.	ChatGPT embeddings
Statistics	Data analysis, model evaluation and pattern discovery.	Fraud detection
Optimization	Finds the best solution by minimizing error or maximizing performance.	Training neural networks
Graph Theory	Used in networks, search engines, recommendation systems and route planning.	Google Maps shortest path
Calculus	Optimizes learning by updating model parameters during training.	Gradient descent

Mathematics in Daily AI Applications

Application	Mathematics Used
Google Maps	Graph Theory + Optimization
Netflix	Probability + Statistics
Amazon	Statistics + Optimization
Face Recognition	Linear Algebra
ChatGPT	Linear Algebra + Probability + Optimization

Spam Detection	Probability + Statistics
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Logic → Reasoning
 Probability → Prediction
 Linear Algebra → Data Representation
 Statistics → Learning from Data
 Optimization → Best Solution
 Graph Theory → Intelligent Search

Activity: Identify one mathematical topic they have already studied and discuss where it could be useful in AI. Encourage examples from Graph Theory, Probability or Linear Algebra.

Research Corner for Interested Students

Explain that modern AI research combines Mathematics with Computer Science. Fields such as Explainable AI, Reinforcement Learning, Optimization, Cryptography and Post-Quantum Security all rely heavily on mathematical foundations.

Quick Quiz

6. Why is Linear Algebra important in AI?
7. Which branch of Mathematics helps AI deal with uncertainty?
8. Which mathematical concept is used in Google Maps?
9. Why is Optimization essential for Machine Learning?
10. True or False: AI can exist without Mathematics.

Homework

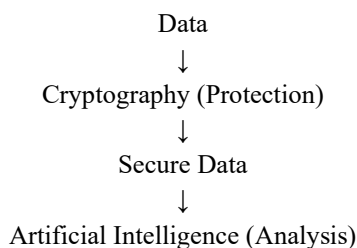
Prepare a one-page chart titled 'Mathematics Behind AI'. For each mathematical subject (Logic, Probability, Linear Algebra, Statistics, Optimization and Graph Theory), write one AI application.

'If Mathematics makes AI possible, how can AI strengthen Cyber Security and Cryptography?' This leads naturally to Part 6: AI and Cryptography.

Lecture 1 – Part 6

AI and Cryptography: Building Secure Intelligent Systems

'In the previous section we learned that Mathematics is the foundation of AI. Now let us see how AI protects our digital world. Every online payment, email, banking transaction and digital signature requires security. This is where Cryptography and Artificial Intelligence work together.'



↓
Intelligent Decision

What is Cryptography?

Cryptography is the science of protecting information from unauthorized access. It ensures confidentiality, integrity, authentication and non-repudiation. Examples include online banking, UPI, digital signatures, HTTPS and secure messaging.

Why Combine AI and Cryptography?

Cryptography protects data, while AI extracts knowledge from data. Modern systems require both. Without cryptography, AI may receive tampered data. Without AI, cyber attacks become harder to detect.

AI in Cyber Security

Application	Role of AI
Spam Detection	Identifies malicious emails using pattern recognition
Fraud Detection	Detects unusual banking or UPI transactions
Intrusion Detection	Recognizes suspicious network activity
Malware Analysis	Classifies malicious software
Phishing Detection	Identifies fake websites and emails
User Authentication	Detects abnormal login behaviour

AI and Digital Signatures

Digital signatures verify the authenticity of digital documents. AI does not replace digital signatures; instead, it complements them by detecting forged signatures, identifying unusual signing behaviour, and helping classify fraudulent documents.

AI vs Cryptography

Aspect	Artificial Intelligence	Cryptography
Primary Goal	Learn and make decisions	Protect information
Input	Data	Plaintext / Keys
Output	Prediction / Decision	Encrypted or authenticated data
Focus	Intelligence	Security
Nature	Adaptive	Mathematical security algorithms

Post-Quantum Cryptography (PQC)

Quantum computers threaten many classical public-key cryptographic systems. Post-Quantum Cryptography develops algorithms that remain secure against quantum attacks. Current NIST-standardized approaches include lattice-based cryptography. AI can assist researchers by optimizing parameter selection, analyzing attacks and discovering implementation anomalies.

Research Corner

Discuss current research topics:

- AI for Intrusion Detection Systems
- AI-assisted Malware Classification
- AI in Fraud Analytics
- Privacy-Preserving Machine Learning
- Federated Learning
- AI for Post-Quantum Cryptography
- Explainable AI for Security

Cryptography → Protects Data
Artificial Intelligence → Learns from Data
Together → Secure Intelligent Systems

Activity

to identify five places where both AI and Cryptography are used together (e.g., UPI, online banking, Aadhaar authentication, e-commerce, secure email). Discuss why both technologies are necessary.

Quick Quiz

11. What is the main purpose of Cryptography?
12. Can AI replace Cryptography? Why or why not?
13. Give two examples of AI in Cyber Security.
14. What is a Digital Signature?
15. What is Post-Quantum Cryptography?

Homework

Prepare a two-page report on 'Applications of Artificial Intelligence in Cyber Security'. Include one recent example related to fraud detection, malware analysis or digital payments.

Bridge to Part 7

'We have explored concepts through discussion. In the next part, we will summarize the entire lecture using flowcharts, mind maps and blackboard diagrams for quick revision.'

Lecture 1 – Part 8

Assessment: MCQs, University Questions, Viva & Self-Evaluation

Section A: Multiple Choice Questions

Q.No.	Question	A/B/C/D (Correct option in text below)	Option Summary	Answer	Bloom
1	Artificial Intelligence is primarily concerned with	A: Making machines perform intelligent tasks B: Operating systems	Concept	A	Remember

		C:Computer networking D:Web design			
2	Which of the following is an example of Narrow AI?	A:AGI B:Google Maps C:Human Brain D:ASI	Concept	B	Remember
3	Which branch of Mathematics is fundamental for route optimization?	A:Graph Theory B:Chemistry C:Biology D:History	Concept	A	Remember
4	ChatGPT is mainly an example of	A:Spreadsheet B:Large Language Model C:Compiler D:Database	Concept	B	Remember
5	AI systems primarily learn from	A:Data B:Luck C:Random guessing D:Electricity	Concept	A	Remember
6	Which one is NOT a characteristic of human intelligence?	A:Reasoning B:Creativity C:Unlimited memory D:Adaptability	Concept	C	Remember
7	Which technology protects information?	A:Cryptography B:Compiler C:HTML D:CSS	Concept	A	Remember
8	Face Unlock mainly uses	A:Computer Vision B:Database C:Sorting D:Compression	Concept	A	Remember
9	Current AI systems are mostly	A:ANI B:AGI C:ASI D:None	Concept	A	Remember
10	Optimization in AI is used to	A:Improve model performance B:Print documents C:Store files D:Increase monitor size	Concept	A	Remember

Section B: Short Answer Questions (2–5 Marks)

16. Define Intelligence.
17. Define Artificial Intelligence.

18. List any five applications of AI.
19. Differentiate Traditional Programming and AI.
20. What is Weak AI?
21. What is AGI?
22. Why is Mathematics important in AI?
23. What is Cryptography?
24. Write any four characteristics of intelligence.
25. Explain AI in one real-life application.

Section C: Long Answer Questions (10–15 Marks)

26. Explain the concept of Artificial Intelligence with suitable examples.
27. Compare Human Intelligence and Artificial Intelligence.
28. Discuss the role of Mathematics in Artificial Intelligence.
29. Explain AI applications in healthcare, education and cyber security.
30. Describe ANI, AGI and ASI with suitable examples.

Section D: Viva Questions

- What inspired the development of AI?
- Can AI think exactly like humans?
- Why is ChatGPT considered AI?
- Is Calculator an AI system? Why?
- What is the difference between learning and memorization?
- How does Google Maps use AI?
- What is the relationship between AI and Cryptography?
- What is Post-Quantum Cryptography?

Expected Viva Discussion Points

Encourage students to answer with examples rather than definitions. Ask follow-up questions such as 'Can you give a real-life example?' or 'Which branch of Mathematics supports this application?'

Self-Evaluation Checklist

- ☐ I can define Artificial Intelligence.
- ☐ I can identify AI applications in daily life.
- ☐ I understand the difference between AI and traditional programming.
- ☐ I can explain ANI, AGI and ASI.
- ☐ I know why Mathematics is important in AI.
- ☐ I can describe one AI application in Cyber Security.